

Beyond Semantic Confidence: Relation-Algebra-Constrained Geometric Compatibility for 3D Scene Graph Relations

Supplementary Material

Anonymous submission

Model and Preprocessing Details

Open3DSG. Open3DSG uses the official non-averaged BLIP checkpoint `epoch=13-step=13104.ckpt`, selected by train/dev validation loss. Unmodified public preprocessing produces candidates for 533/548 contexts; the main evaluation retains the full 548-context ground-truth universe and assigns no predictions to the 15 drops. For sensitivity, the eligible-only denominator contains 3,899 relations, and a documented recovery covers all 548 contexts by lowering the visible-object minimum to two and regenerating views for two scans. The release manifest records all three routes.

SGFN. We use the official `full_1160` source with its 160-object, 26-relation checkpoint. The evaluated scans, denominator, methods, and tensor-shape checks are recorded in the supplemental package.

Compatibility and Structural Diagnostics

RelCompat3D fits one logistic model per measured relation family. Numeric features use training-only means and standard deviations; missing values use the corresponding training mean. Inputs are predicate/family indicators T , predicate-independent pair geometry G , and explicit $T \times G$ interactions. Source score, source rank, source identity, and object class are forbidden. Orbit-consistent augmentation preserves the proximity predicate under endpoint swap; vertical order swaps endpoints and applies the inverse predicate. Linked counterfactuals receive the fixed margin loss in the main paper. Support/contact receives the pairwise loss but no blanket endpoint transform.

The 60,208 training rows yield 33,961 linked pairs, and algebra augmentation produces 86,032 optimization rows. We use 800 deterministic full-batch steps at learning rate 0.2; the three family models contain 22–24 parameters. The support/contact model is used only in unrestricted-fusion diagnostics, while the primary family-slot route preserves that family’s source ordering.

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Table 1: **Training and development discrimination.** Labels comprise GT-positive/counterfactual pairs, not human physical-validity judgments.

Family	Train	Dev	Param.	Train pairs	AUROC	Brier
Proximity	20,902	2,106	22	9,562	1.0000	.0009
Vertical order	4,922	566	23	1,644	.9982	.0079
Support/contact	34,384	3,574	24	22,755	.9933	.0256

Across all 6,246 development rows, AUROC/AUPRC/Brier are .9973/.9973/.0157. On 3,516 linked development pairs, the positive wins 99.23% of the time with mean logit margin 14.13. Correct-predicate compatibility wins on 97.40% of 384 vertical positives. Exact orbit projection gives maximum proximity-swap and vertical-inverse errors of zero on development and all three final source outputs.

The factor-isolation ablation uses the same split and estimator class. At development, T -only, true raw- G -only, additive $T + G$, and interaction $T \times G$ obtain AUROC .6124/.8809/.9193/.9822 and Brier .2370/.1363/.1061/.0495. Shuffled and mismatched-pair geometry reduce the intended reliability signal. These results verify factor use and the implemented relation algebra; they do not convert the constructed target into independent physical-validity ground truth.

Table 2: **Unrestricted-fusion mechanism audit at $K = 100$.** Cells are Recall / verifier V. Unlike the primary routed ablation, compatibility is applied to every family; Open3DSG uses the documented full-coverage sensitivity so each corruption has a donor in every context.

Condition	VL-SAT	Open3DSG	SGFN
Structured product	.9688/.0325	.6055/.0339	.9418/.0372
Wrong predicate	.9507/.0745	.5753/.2315	.9182/.1120
Wrong-pair G	.9121/.0525	.2885/.0891	.8429/.0723
Shuffled G	.9089/.0615	.3326/.1418	.8338/.0861
Label-fixed swap	.9509/.0745	.5775/.2325	.9187/.1120
Distance only	.5554/.0981	.5038/.1071	.5554/.0981
Compatibility only	.6186/.0193	.5798/.0321	.6186/.0193

Strong-Comparator and Open3DSG Sensitivities

Supervision-matched nonlinearity. To separate architecture capacity from supervision, we fit a 69-parameter two-hidden-unit MLP using the same 60,208 training rows, constructed compatibility target, feature contract, source-score exclusion, and 1,061/117/157 split as the linear model. It is fitted once and applied unchanged to all three predictors. Table 3 also shows a separate nonlinear rescorer trained on stronger SGFN-specific exact-label correctness. Open3DSG uses the complete-coverage sensitivity here so every comparator sees the same candidate set.

Table 3: **Nonlinear comparison at $K = 100$.** Cells are Recall / verifier V. The matched MLP does not dominate the linear structured product; the last column uses stronger SGFN-specific supervision.

Source	Linear product	Matched MLP	Exact-label MLP
VL-SAT	.9688/.0325	.9688/.0317	.9625/.0311
Open3DSG	.6055/.0339	.6415/.0434	.6166/.0334
SGFN	.9418/.0372	.9421/.0363	.9466/.0279

Against the linear product, the matched MLP’s paired Open3DSG changes are $+0.0360$ Recall $[.0252, .0475]$ and $+0.0096$ V $[.0079, .0113]$. On VL-SAT and SGFN, Recall differences are within ± 0.0003 and V differences are at most $.0010$ in magnitude for this structured MLP. The stronger exact-label model is best on its SGFN source but is worse than routed RelCompat3D on both VL-SAT metrics and exhibits a Recall–V trade-off on Open3DSG. This supports the factorized compatibility mechanism, not universal formula superiority.

Open3DSG public-route sensitivity. The public preprocessing drops a context when fewer than four annotated objects retain view metadata. This accounts for all 15 missing contexts. Table 4 distinguishes the faithful eligible-context evaluation, the conservative full-target main evaluation, and a documented full-coverage recovery.

Table 4: **Open3DSG route sensitivity at $K = 100$.** Cells are Recall / verifier V. The main route uses the full 3,972-relation denominator and no predictions for the 15 public-pipeline drops.

Route	Contexts / GT	Source	Routed RelCompat3D
Public eligible	533 / 3,899	.5206/.1242	.5799/.0324
Public, full target (main)	548 / 3,972	.5111/.1242	.5692/.0324
Recovered full coverage	548 / 3,972	.5161/.1242	.5743/.0332

The conclusion is stable to all three denominator and coverage choices. The main paper uses the conservative full-target public route; recovered results are sensitivity evidence rather than the headline benchmark.

Primary Scan-Cluster Confidence Intervals

The primary analysis resamples the 157 scan identifiers 1,000 times and carries every context of a sampled scan together. The ranking, model, and point estimates remain unchanged; shared bootstrap indices are used across conditions. A paired 548-context bootstrap gives the same directions and is retained as a finer-grained sensitivity.

Table 5: **Primary scan-cluster intervals at $K = 100$.** Entries are routed RelCompat3D minus source score with paired 95% intervals.

Source	Δ Recall	Δ Verifier-V
VL-SAT	+0.023 $[.0005, .0049]$	−.0182 $[−.0199, −.0166]$
Open3DSG	+0.0582 $[.0470, .0695]$	−.0917 $[−.0963, −.0873]$
SGFN	+0.0068 $[.0043, .0098]$	−.0280 $[−.0304, −.0257]$

All Verifier-V intervals remain below zero and all Recall intervals remain above zero despite the near-saturated VL-SAT baseline.

Family Composition at the Global $K = 100$ Operating Point

Table 6 reports each family’s slice inside the actual global top-100 ranking. Deltas compare routed RelCompat3D with the source score using shared paired context-bootstrap indices.

Table 6: **Global-top-100 family slices.** Positive ΔV is a regression. Values are verifier-derived.

Source	Family	ΔR	ΔV
VL-SAT	Support/contact	.0000	.0000
VL-SAT	Proximity	+0.0040	−.0076
VL-SAT	Vertical order	+0.0051	−.0609
Open3DSG	Support/contact	.0000	.0000
Open3DSG	Proximity	+0.0968	−.0443
Open3DSG	Vertical order	+0.1538	−.3014
SGFN	Support/contact	.0000	.0000
SGFN	Proximity	+0.0142	−.0150
SGFN	Vertical order	+0.0051	−.0643

Support/contact is exactly unchanged because the route preserves both its selection and the source-ranked family composition at every K . Vertical order drives most of the aggregate reduction. The result is consequently a scoped operating-point improvement, not a uniform family claim. Full within-family and global-slice results for $K \in \{5, 10, 20, 50, 100\}$, including simultaneous family-wise intervals, are included in the supplemental package.

Relative-Size Scope Extension

We evaluate bigger than/smaller than as secondary breadth evidence, not as core evidence that learned compatibility is necessary. The extension uses the same 1,061/117/157 train/development/evaluation split. Its model

receives predicate sign, robust point extents and volume ratios, and signed predicate–geometry interactions; source score, rank, object class, and evaluation labels are excluded. Alternating point subset A (p05–p95 extents) provides model evidence, whereas disjoint subset B (p02–p98 extents) provides the verifier measurement. Annotation OBB volume is used only by a fixed-rule baseline.

The model fits 3,244 training rows and obtains target-discrimination AUROC .9974 on 376 development rows. Correct-predicate ordering is 96.81%; exact inverse-equivariance and common-scale invariance have maximum error 2.22×10^{-16} . Table 7 summarizes the $K = 100$ result; the supplemental package contains all five K values and family compositions.

Table 7: **Relative-size extension at $K = 100$.** Within-size Recall is unchanged for all sources. Global columns use the four-family denominator of 4,142; parentheses are paired 95% context-bootstrap intervals.

Source	Within-size ΔV	Global ΔR	Global ΔV
VL-SAT	−.1025 [−.1063, −.0988]	+.0048 [.0020, .0080]	−.0239 [−.0261, −.0218]
Open3DSG	−.1274 [−.1383, −.1167]	+.0698 [.0478, .0935]	−.1040 [−.1090, −.0997]
SGFN	−.0593 [−.0616, −.0572]	+.0256 [.0199, .0317]	−.0359 [−.0381, −.0336]

The fixed robust-point rule is as strong or stronger than the learned model on within-family violation: learned-minus-rule ΔV is +.00318 for VL-SAT, +.00005 for Open3DSG, and +.00367 for SGFN. Disjoint point subsets remove exact rule reuse but not shared-geometry construct dependence. The family therefore broadens the measured scope while remaining a rule-easy secondary analysis.

Verifier-Uncertainty Sensitivity

The primary $V@K_{all}$ includes uncertain rows in its denominator but does not label them satisfied. We therefore recompute each ranking with $V@K_{dec}$ over satisfied/violated rows, uncertainty rate $U@K$, and a pessimistic bound $V@K_{pess}$ that counts every uncertain row as a violation. Table 8 gives the primary $K = 100$ comparison.

Table 8: **Uncertainty sensitivity at $K = 100$.** Lower is better for all four reported quantities.

Source	Condition	V_{all}	V_{dec}	U	V_{pess}
VL-SAT	Source score	.0476	.0729	.3464	.3941
VL-SAT	RelCompat3D routed	.0295	.0452	.3473	.3768
Open3DSG	Source score	.1242	.2126	.4161	.5402
Open3DSG	RelCompat3D routed	.0324	.0548	.4084	.4408
SGFN	Source score	.0630	.1005	.3732	.4362
SGFN	RelCompat3D routed	.0350	.0565	.3803	.4153

Routed-minus-source changes in V_{dec} are −.0277 for VL-SAT, −.1578 for Open3DSG, and −.0440 for SGFN; their paired 95% intervals all exclude zero. Uncertainty-rate changes are +.0009, −.0077, and +.0071, whereas pessimistic-V changes are −.0173, −.0994, and −.0209.

Thus the primary conclusion holds without requiring uncertainty to decrease uniformly. The hard-rule baseline has zero all-status V but retains uncertain rows and therefore has pessimistic V of .3712/.4722/.4042. Thus zero measured violation does not imply reliable or complete selection.